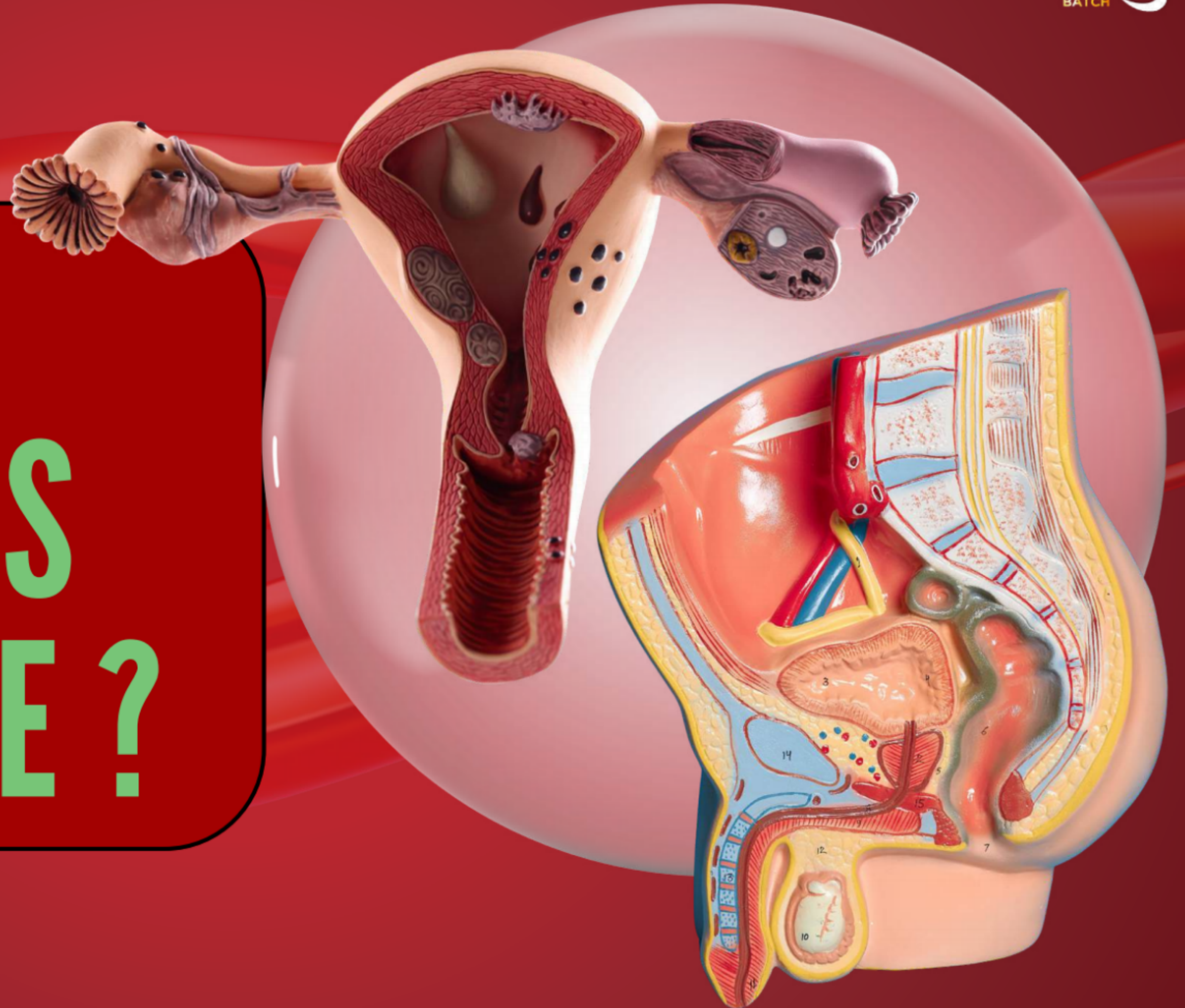


L4

CLASS X - SCIENCE

HOW DO ORGANISMS REPRODUCE ?

PRASHANT KIRAD



Check yourself

Q. Two identical flowers P and Q are taken. In P, the stigma is covered before pollen grains reach it. Q is left uncovered. Which flower is more likely to form seeds? Explain.

Answer: Flower Q is more likely to form seeds.

In flower P, the stigma is covered, so pollen grains cannot reach it and pollination does not occur. Therefore, fertilisation cannot take place and seeds will not form.

- In flower Q, the stigma is exposed, so pollen grains can land on it. The pollen grain can then germinate, form a pollen tube, and carry the male gamete to the ovule for fertilisation, leading to seed formation.

Check yourself

Ques: A flowering plant has two flowers, A and B. Pollen grains from the anther of flower A are transferred to the stigma of flower B. In another situation, the pollen from the anther reaches the stigma of the same flower.

1. Name the type of pollination in both flowers.
2. Mention two agents that can bring about the transfer of pollen between flowers?
3. Explain why pollen must reach stigma before fertilization can occur
4. What structure develops from pollen grain to carry the male germ cell towards the female germ cell?

1. Flower A → Flower B: Cross-pollination

Same flower: Self-pollination

2. Two agents of pollination: Wind and Insects

3. Pollen must reach the stigma because after landing on the stigma, the pollen grain germinates and forms a pollen tube. This tube carries the male gamete towards the ovule, where fertilisation occurs.

4. The structure is the pollen tube. It develops from the pollen grain and carries the male germ cell towards the female germ cell.

Check yourself

Q. Three similar flowers are subjected to the following treatments: in flower A, stamens are removed before the anthers mature; in flower B, the stigma is removed before pollination; and in flower C, the ovary is removed after pollination but before fertilisation. Predict how each treatment would affect pollination, fertilisation and the formation of seeds/fruits. Give reasons for your answer.

- **Flower A – Stamens removed before anthers mature:**

Self-pollination will be prevented because the flower cannot produce its own pollen. However, cross-pollination can still occur if pollen from another flower reaches its stigma. If fertilisation occurs, seeds and fruit can still form. This follows from the lecture's distinction between self- and cross-pollination.

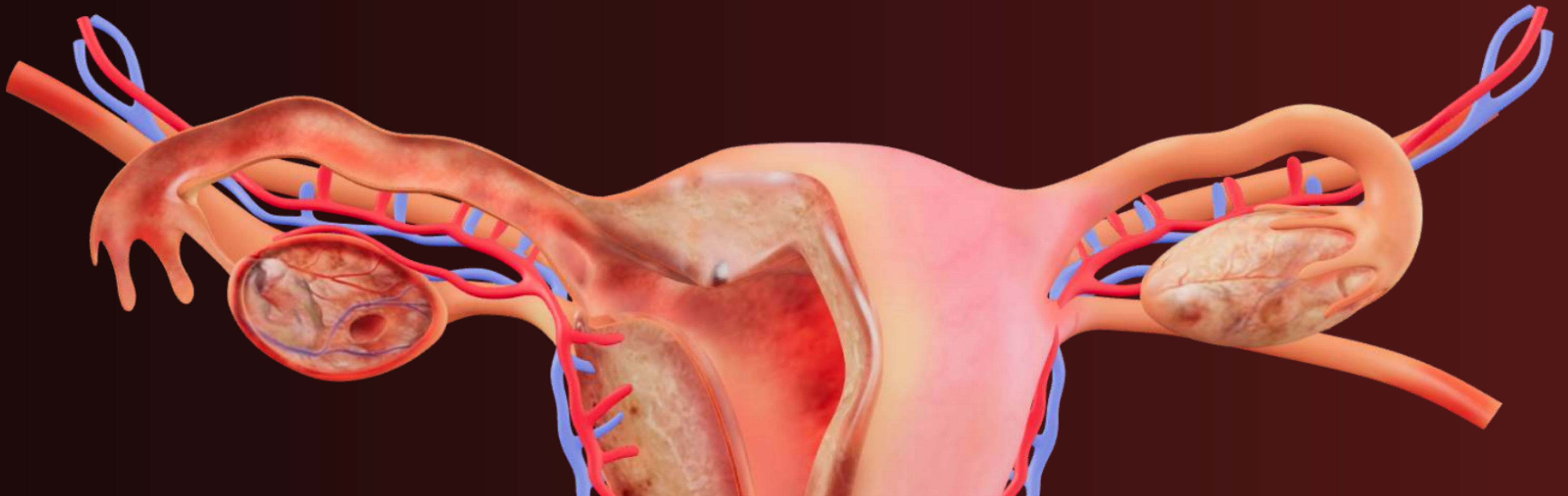
- **Flower B – Stigma removed before pollination:**

Pollination will not occur because the stigma is the part that receives pollen grains. Therefore, fertilisation will not take place and seeds and fruit will not form.

- **Flower C – Ovary removed after pollination but before fertilisation:**

Fertilisation cannot occur because the ovules containing the female gamete are present inside the ovary. Hence, seeds will not form. Also, since the ovary develops into the fruit after fertilisation, fruit formation will also not occur.

HUMAN REPRODUCTIVE SYSTEM



Male
(Sperm)



Female
(Ovum)

Puberty

Q. What is puberty? Mention any two changes common to boys and girls during puberty.

Puberty is a time period or stage of development during which hormonal changes bring about sexual maturation and the development of secondary sexual characteristics.

- It marks the beginning of adolescence, when both physical and hormonal changes occur in boys and girls.

What happen during PUBERTY?

- During puberty a person undergoes physical, emotional and mental changes.

Reproductive Glands



• Common Changes in both male and female

- Growth of thick hair in armpits & genital area.
- Thinner hair on arms, legs, face.
- Skin becomes oily → pimples.
- Height and body proportions change.
- Awareness of one's own body & others increases
- Changes in voice



Changes in boys

- New facial hair growth
- Testes begin producing sperms (spermatogenesis)
- Penis enlarges, erection possible



Changes in Girls

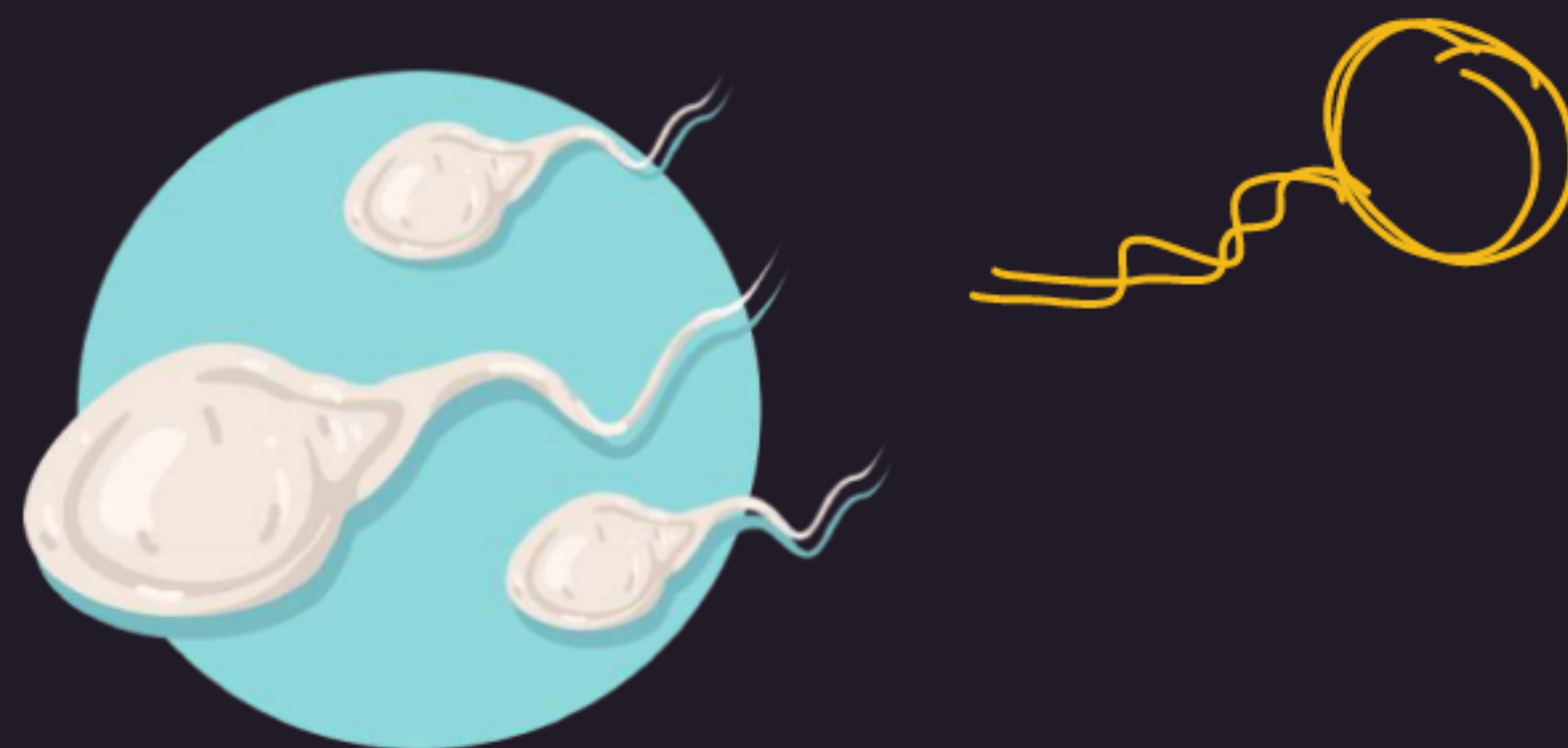
- Increase in breast size
- Menstruation (Menarche)
- Ovary begin producing egg cells (oogenesis)

Check yourself

Question: A boy reaches puberty and shows visible physical changes. Which statement best explains the biological significance of these changes?

- a) They indicate that general body growth has stopped.
- ☒ b) They are signs associated with sexual maturation.
- c) They indicate that reproductive organs are no longer developing.
- d) They occur only after fertilization

Male Germ-Cell (Sperm)	Female Germ-Cell (Egg/Ovum)
Produced in testes .	Produced in ovaries . 
Very small , motile (can move).	Large, non-motile (cannot move).
Millions are produced continuously after puberty.	✓ Usually one ovum released in a month after puberty.
Motile with a tail (flagellum) for movement.	Round in shape, no tail.



Male Reproductive System

Primary sex organs – Testes

- Produce **sperms** (male germ cells).
- Secrete the hormone **testosterone**.
- Testosterone regulates sperm formation and causes changes seen in boys at puberty.

Why are testes in the scrotum?

- Sperm formation requires a temperature lower than normal body temperature.

Path of sperms: **Testis** → **Vas deferens** → **Urethra** → **Outside**

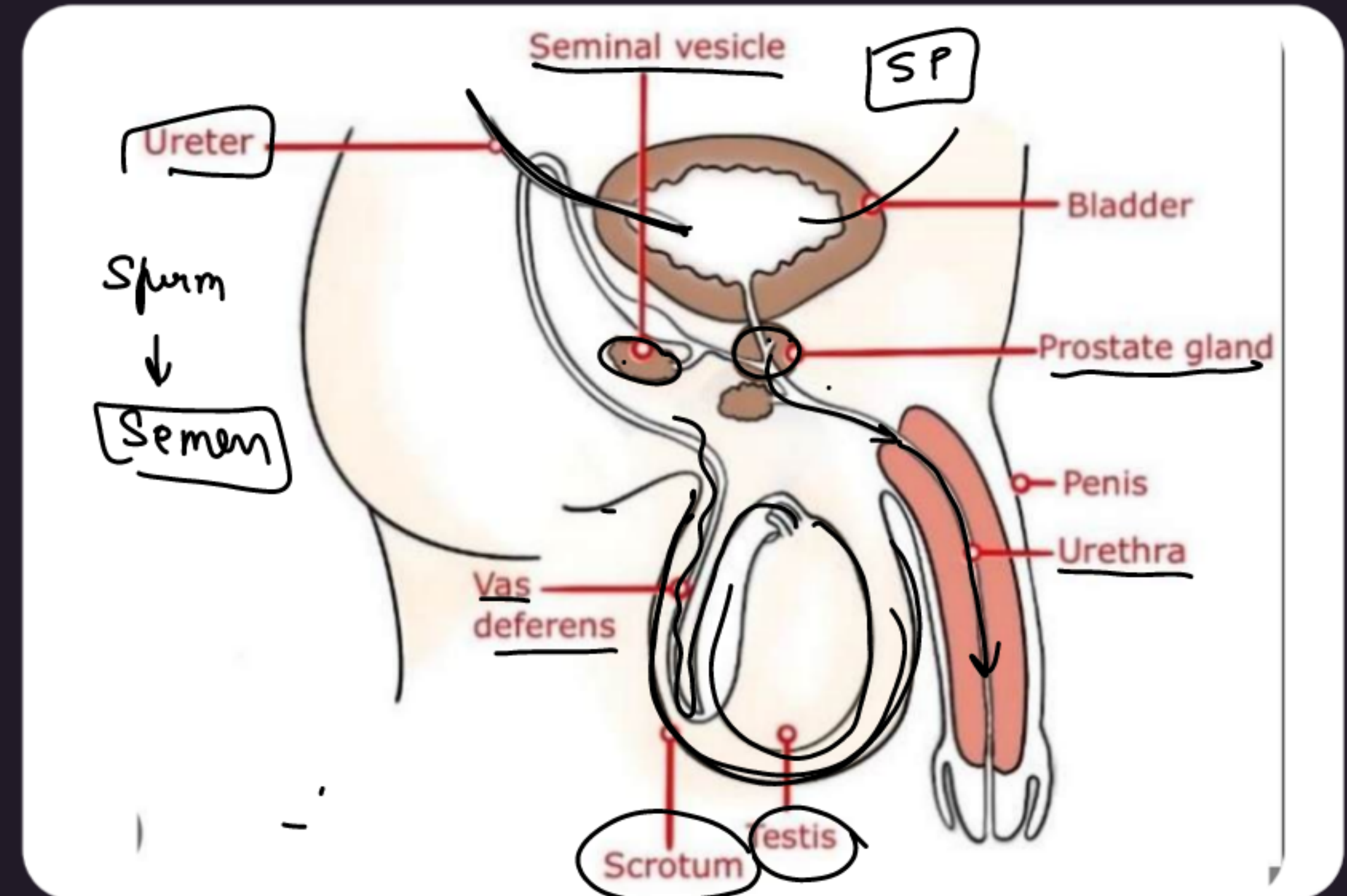
④ Role of seminal vesicles and prostate gland

- Add secretions to sperms. → *Flow*
- This fluid makes transport easier and provides nutrition.

Urethra: Common passage for **urine** and **sperms**.

A Sperm (Sperm Cell)

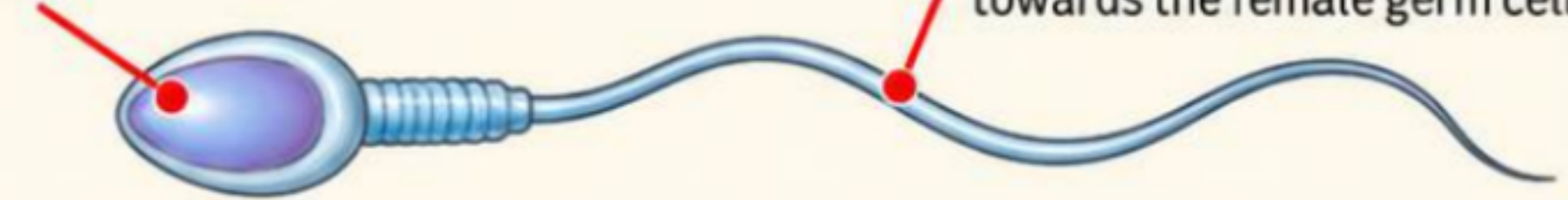
- Head: genetic material
- Tail: helps in movement towards the female germ cell



A Sperm (Sperm Cell)

Head: genetic material

Tail: helps in movement towards the female germ cell



Organ	Location / Structure	Function
Testes	Located in the <u>scrotum</u> , outside the <u>abdominal cavity</u> .	Produce sperms and secrete testosterone (controls puberty changes in boys).
Scrotum	<u>Sac-like structure around testes.</u>	Keeps testes at a lower temperature than body for sperm production.
Vas deferens	<u>Tube carrying sperms from testes.</u>	Transports sperms towards urethra.
Seminal vesicle	Along vas deferens, near bladder.	<u>Adds fluid to sperms</u> → provides nutrition and helps transport.
Prostate gland	Along vas deferens, near urethra.	<u>Adds fluid to sperms</u> → makes movement <u>easier</u> .
Urethra	Tube that runs through penis.	Common passage for sperms and urine .
Penis	External organ.	Transfers sperms into the female body.

Q. What are accessory glands in the male reproductive system? Write their functions.

The accessory glands include seminal vesicles and the prostate gland.

Functions:

1. Seminal vesicles add a sugar-rich fluid that provides energy to sperms.
2. Prostate gland secretes a milky fluid that increases sperm mobility and helps in their transport.
→ Together, these secretions mix with sperms to form semen.

Q. Why are testes located outside the abdominal cavity in scrotum?

Testes are located in the scrotum outside the body to maintain a temperature $2-2.5^{\circ}\text{C}$ lower than body temperature, which is essential for sperm production and maturation.

Sperm +

Q. What is semen? What is its function?

- Semen is a mixture of sperms and secretions from seminal vesicles and prostate gland.

Function: It provides nutrition, energy, and fluid medium for sperms to swim and reach the egg for fertilisation.

Check yourself

Question: A student is studying the male reproductive system and make three observations—

1. Sperm are produced in one organ.
2. Glands add secretion to the sperm.
3. The resulting fluid is easier to transport and also provides nutrition.

- a) Name the organ where sperm are produced? ✓
- b) Name two glands mentioned in this process.
- c) Why are their secretions important? ✓

a) Testes

b) Two glands involved: Seminal vesicles and Prostate gland

c) Their secretions provide nutrition to sperms and make their transport easier by forming a fluid medium

Check yourself

- Q. (a) What are the functions performed by the testes in human beings?
 (b) State the role of seminal vesicles and prostate gland in the male reproductive system.
 (c) Why does a sperm have a long tail?

(a) Functions of testes in human beings:

- Produce sperms.
- Secrete the male sex hormone testosterone, which brings about changes at puberty and helps regulate sperm formation.

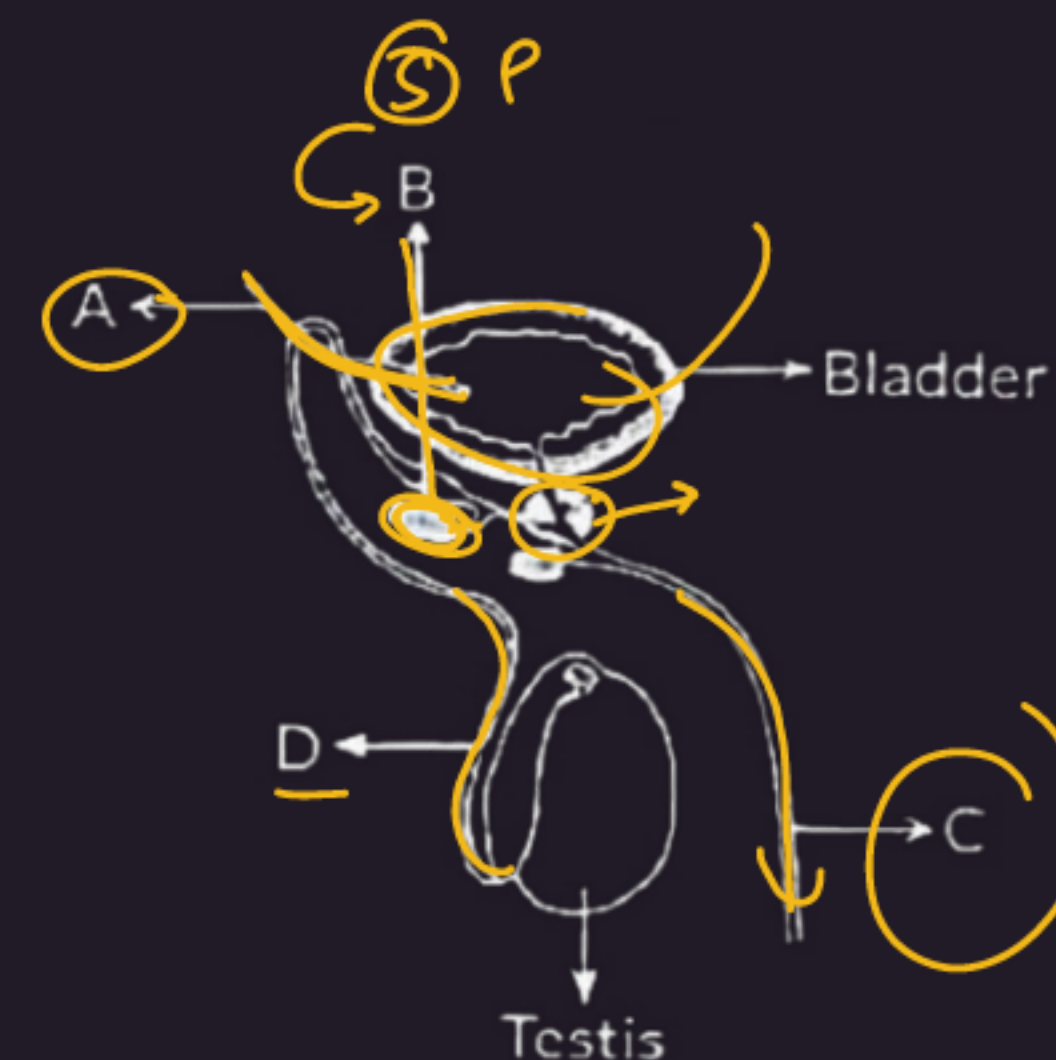
(b) They add secretions to the sperms. These secretions provide nutrition and help in easier transport/movement of sperms.

(c) The long tail helps the sperm move/swim towards the female germ cell (egg).

Check yourself

Q. Based on the given diagram, answer the questions given below:

- (a) Label the parts A, B, C, and D.
- (b) Name the hormone secreted by testis and mention its role.
- (c) State the functions of B and C in the process of reproduction.



Ans: (a) A-Ureter

B - Seminal vesicle

C - Urethra

D - Vas deferens

(b) Testes produce male sex hormone testosterone. Hormone testosterone brings about the development of secondary sexual characters during puberty in boys like growth of facial hair, deepening of voice, build up of muscle mass and also regulates formation of sperms.

(c) Seminal vesicles (B) release its contents into the ejaculatory duct during ejaculation.

Urethra (C) carries sperms from the vas deferens through the penis.

Female Reproductive System

Primary sex organs – Ovaries

- Produce eggs (female germ cells / ova).
- Also produce female sex hormones (oestrogen and progesterone).

Egg release

- At puberty, some eggs begin to mature.
- One egg is released every month by one ovary (ovulation).

Path of egg: Ovary → Oviduct / Fallopian tube → Uterus

Role of oviduct / fallopian tube

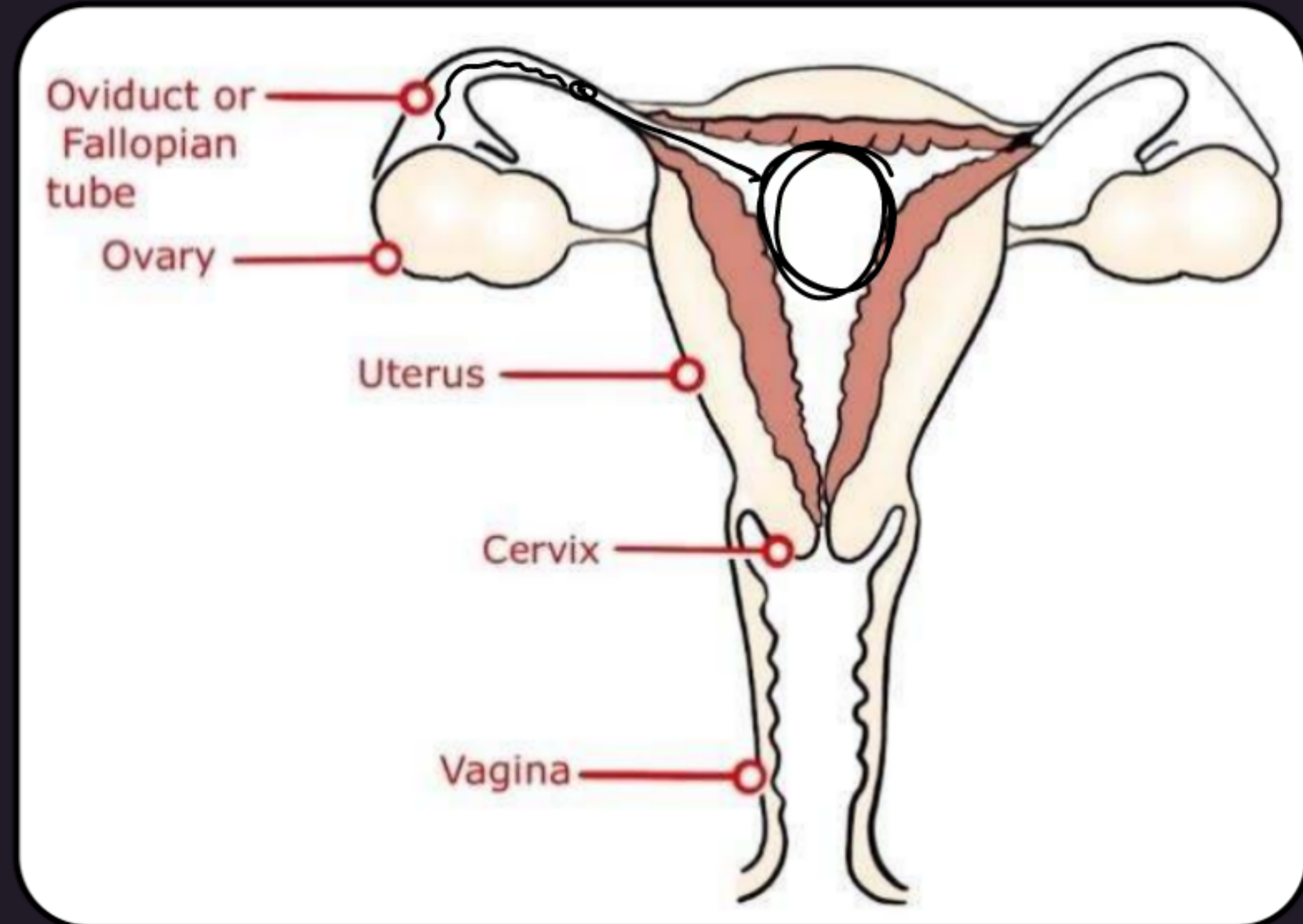
- Carries the egg from ovary to uterus.
- Fertilisation usually occurs here.

Role of uterus

- Muscular / elastic bag-like organ.
- Receives the fertilised egg.
- Embryo develops here.

Cervix and vagina

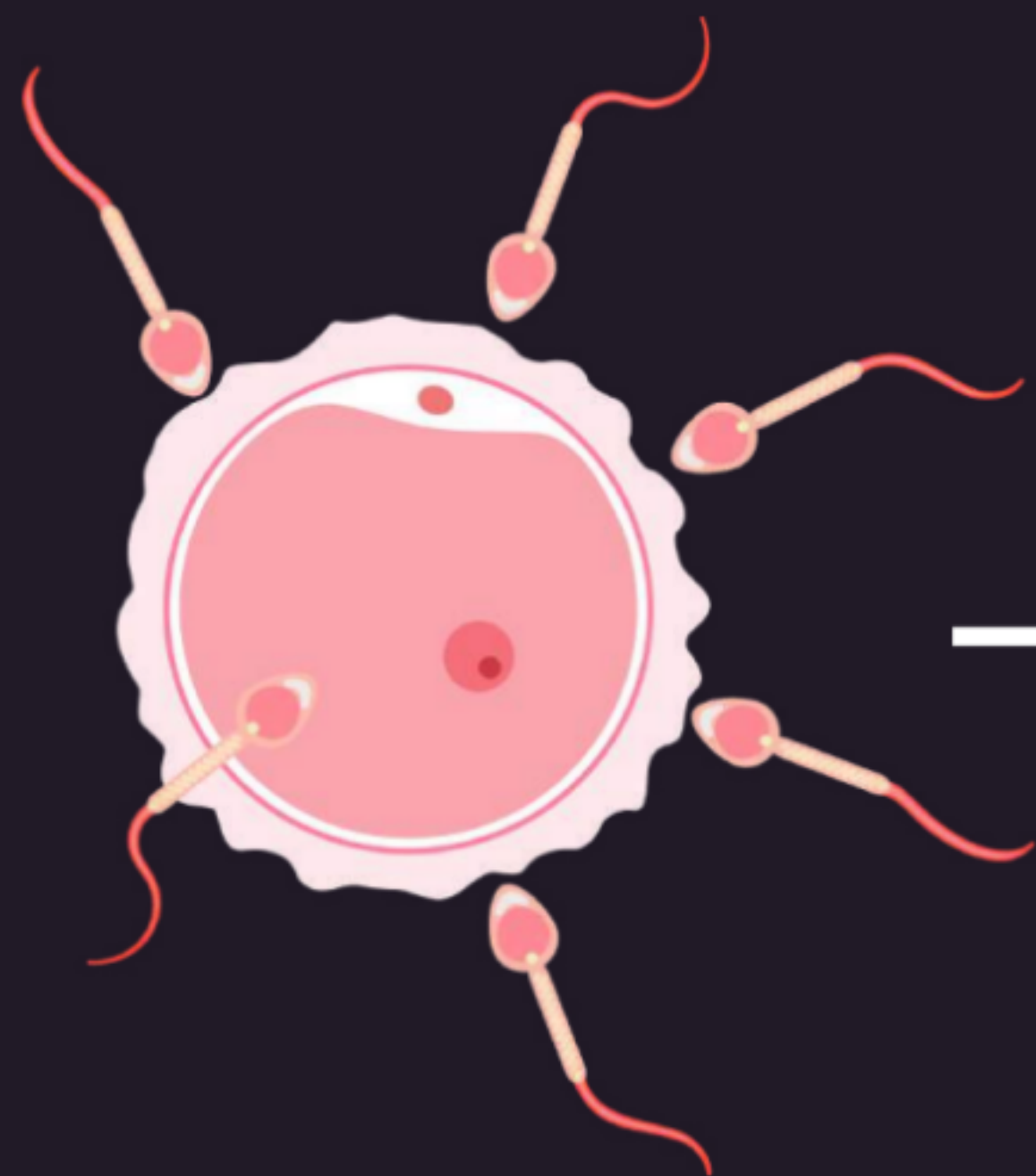
- Uterus opens into the vagina through the cervix.
- Sperms enter through the vaginal passage.



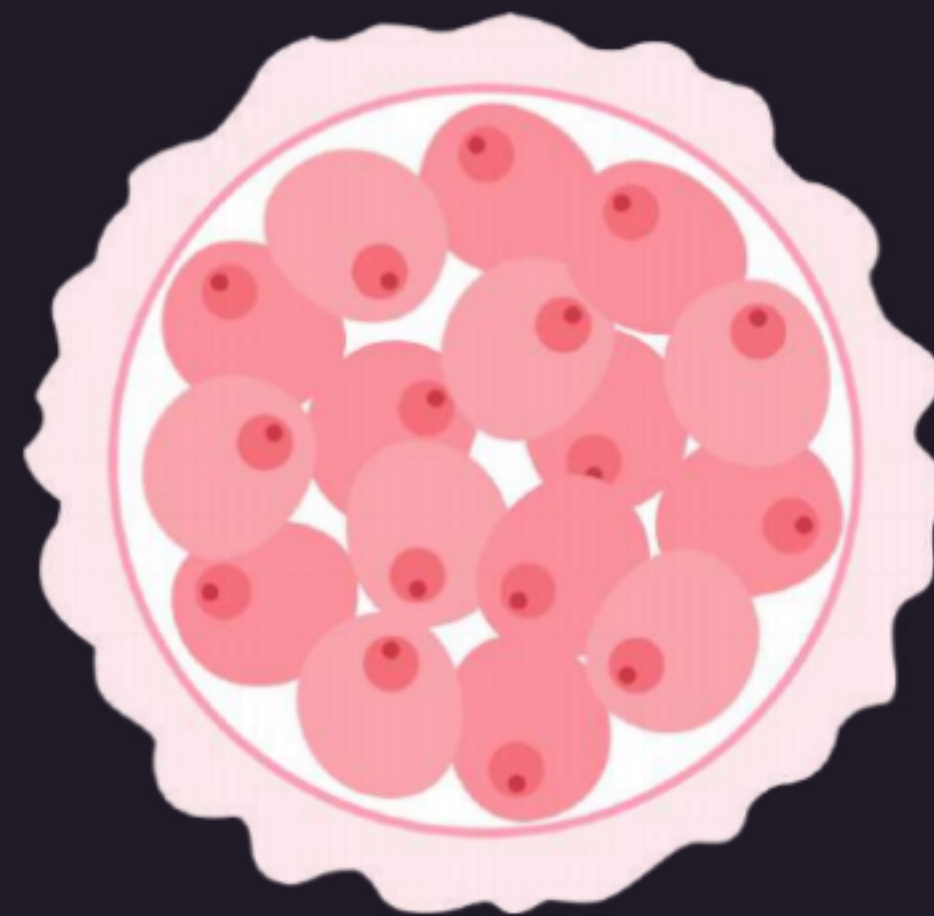
If fertilisation occurs:

- Zygote forms.
- Embryo gets implanted in the uterus.
- Embryo develops into a fetus.

Organ	Location / Structure	Function
Ovary	Paired organs in the lower abdomen, one on each side of uterus.	Produce female gametes (ova/eggs) and secrete hormones (estrogen, progesterone) .
Ovulation	Process in the ovaries, once every month.	Release of a mature ovum from an ovary.
Fallopian Tubes (Oviducts)	Thin tubes extending from ovaries to uterus.	Carry egg from ovary to uterus; site of fertilisation .
Uterus (Womb)	Hollow, muscular organ in pelvic cavity.	Embryo implants in uterine lining (endometrium); supports development into foetus .
Cervix	Narrow, lower part of uterus.	Connects uterus to vagina; passage for sperms; dilates at childbirth .
Vagina	Muscular canal from cervix to outside of body.	Receives sperms during intercourse; acts as birth canal at delivery.



Zygote



Embryo



Foetus



Baby